Jake Feldman Starosta

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Skills

Software: Python, C/C++, Java, MATLAB, JavaScript, HTML/CSS, SQL
Hardware: LTSpice, Arduino, VHDL, SolidWorks

Education

BEng in Applied Mathematics and Computer Engineering – Queen's University — Sept 2023 - April 2027E

4.0/4.3 GPA, Dean's List (all years)

Relevant courses: Electronics, Computer Architecture, Digital Systems, Microprocessors, Algorithms, Software Development, Data Structures, Databases, Control Theory

Professional Experience

Undergraduate Research Assistant – *Queen's University* — May 2026 - Present

- Conducting NSERC-Funded research on stochastic control and reinforcement learning under Dr. Serdar Yüksel

Computer Engineering Intern – *Corecurrent Solutions* — May 2025 - August 2025

- Reverse-engineered PCBs via wire tracing, chip identification, and patent analysis to detect IP infringement
- Produced litigation-quality technical reports
- Pitched and implemented a tool that reduced IC identification time by 60% using vectorized search and a classification model achieving 94% manufacturer prediction accuracy

Full-Stack Developer – *EduTutor* — Jan 2025 - April 2025

- Contributed to an LLM-powered platform that generates personalized tutor curricula from course material
- Shipped an AI-generated content quiz editing feature for students

Workshop Tutor – *EngLinks* — Sept 2024 - April 2025

- Taught calculus tutorials to 250 engineering students at Queen's

Extracurricular Activities

Conference Chair – *Canadian Undergraduate Conference on AI* — May 2025 - April 2026

- Led a team of 12 volunteers across 4 universities, managing a $60k operating budget
- Delivered the largest conference to date: 400+ attendees, with partners including NVIDIA, Shopify, KPMG, Botpress, CIFAR, and Anthropic

Director of Design – *QMIND: Queen's AI* — May 2025 - April 2026

- Oversaw three AI projects: an automated medical-imaging triage system, an embedded neural network on a mobile robot and a retrieval-augmented generation consulting project

Project Manager – *QMIND: Queen's AI* — April 2024 - April 2025

- Managed a team of 6 undergraduate AI researchers to design a novel symbolic music genre transfer architecture
- Published in CUCAI 2025 proceedings and presented our work at the Toronto AWS office

Projects

ClarityAI: Speech-to-Speech Accessibility Platform

- Real-time speech pipeline that transcribes, rewrites, and respeaks for ESL and neurodivergent accessibility
- Won the ElevenLabs award track at QHacks 2026 (across 60+ teams)

Autonomous Drone (In Progress)

- Implementing a PID flight controller from scratch on an ESP32 for a custom-built quadcopter
- Integrating an NVIDIA Jetson for onboard object detection